Flash Endurance Engineering for Write-Intensive Embedded Workloads: Extending SD Card Lifespan Through Kernel-Level Optimization

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Abstract—Low-cost flash has become the default storage medium for many embedded devices to date. Meanwhile, the endurance limits of flash memory have been a major reliability concern whenever such devices need to handle frequent writes, such as during intensive machine-learning workloads. Default linux filesystems generate a high number of small writes which are detrimental to flash storage that typically comes with little over-provisioning, thus significantly limiting the lifespan of such devices. We have developed an algorithm which dramatically improves the endurance of small random writes.

To address this reliability feature, this experiment was conducted to evaluate the storage characteristics of a representative 32Gb SD card. A set of coordinated kernel- and filesystem-level optimizations have been applied to reduce write amplification. With the help of such technologies as swap compression (ZRAM), page-cache consolidation, multi-queue NOOP scheduler, and log-structured F2FS filesystem, the write amplification factor (WAF) has been reduced from 12.43 to only 2.10. As a result, daily writes were decreased from 4.2Gb to approximately 0.8Gb (Fig. 1).

When applied in the context of a JEDEC-endurance projection model, these reductions lead to a storage system lifetime projection that is three orders of magnitude higher than what was initially measured under the same JEDEC workload. Moreover, the optimized storage stack dramatically reduces tail-latency spikes due to intensive garbage collection in aged filesystems. The results illustrate benefits of host-side I/O shaping for prolonging the service lifetime of low-cost flash storage. The improvements are especially valuable in the context of continuous embedded operation.

Index Terms—Flash Endurance Engineering, Write Amplification Factor, Log-Structured Filesystem, JEDEC Endurance Projection, Kernel Writeback, Embedded Storage Reliability.

I. Introduction

The main argument of this paper is that host-side I/O shaping can greatly reduce write amplification and improve flash endurance in embedded edge systems. Edge computing systems often leverage cheap forms of flash storage as a way to reduce costs and make hardware replacements easier. One such example is the Secure Digital, or SD, cards that are widely used due to their affordability and prevalence.

A. Why Continuous Edge Telemetry Accelerates NAND Wear

Consumer SD cards are optimized for burst-oriented usage patterns—intermittent photo capture, sequential video recording, and occasional firmware updates. Embedded AI workloads violate these assumptions fundamentally. Continuous telemetry logging, container metadata churn, system swap activity, and routine OS updates generate persistent small-write patterns that aggressively stress low over-provisioned flash controllers [19]. Metadata-heavy telemetry can cause FTL GC to do extra work: for each additional 4 KB JSON message appended, the controller must internally read-modify-write whole multi-megabyte erase blocks. Low over-provisioning ($\alpha \approx 0.05$) increases this effect since the controller has fewer spare blocks to choose from when performing garbage collection. The result is that workload characteristics, not storage capacity, dictate endurance [9].

In resource-constrained edge nodes operating with limited RAM, no UPS, and demanding continuous uptime, the interaction between OS write behavior and flash controller becomes the dominant durability factor [24].

B. Contributions and Subsystem Scope

Rather than modifying proprietary flash-controller firmware, this work explores whether host-side operating-system optimizations alone can mitigate this endurance challenge. Our approach relies on four components:

1) Write Amplification Characterization: We calculate the WAF and GC cost incurred by continuous telemetry workloads on embedded platforms.
2) Kernel-Level I/O Shaping: We manipulate the Linux VFS page-cache state machine to force write-clustering, supplemented by ZRAM-based swap compression and I/O scheduler tuning.
3) Filesystem Architecture Comparison: We evaluate F2FS against Ext4, demonstrating how eliminating the JBD2 journal penalty reduces physical amplification.
4) JEDEC-Aligned Lifespan Modeling: We ground endurance projections using empirical kernel block statistics captured via eBPF tracing.

Within the ScholarMaster architecture, this subsystem performs flash-endurance optimization, host I/O shaping, and write-amplification reduction only. It does not orchestrate deployment infrastructure (Paper 20), validate run-

time privacy enforcement (Paper 18), optimize federated adaptation (Paper 13), derive formal verification theorems (Paper 21), or define architectural privacy doctrine (Paper 17).

II. NAND Flash Mechanics and FTL Modeling

Understanding the endurance limitations of SD cards requires examining the physical structure of NAND flash memory. This structure directly constrains write endurance.

A. The Physics of Wear and Page/Block Asymmetry

NAND flash memory has a hierarchical geometry of storage arrays. Inside the flash memory, pages (typically 4KB to 16KB) are grouped into blocks (typically 128 to 256 pages). One of the main challenges associated with NAND Flash memories is that while data can be programmed at the finest level of page granularity (typically 4 KiB – 16 KiB), it can only be erased at the block level [5].

Because pages cannot be overwritten, any change, however small, requires that the controller write new data to a different, erased page and then mark the original page as invalid.

At a physical level, wear occurs due to the continuous program/read cycles that put stress on the tunnel oxide in between the floating gate. The Fowler-Nordheim tunneling mechanism that occurs during the erase phase results in oxide degradation when program/erase cycles are applied repeatedly on the cell. The distribution of threshold voltages of the cell gets shifted as the floating gate accumulates charges, thus reducing the amount of margins required for reading. This creates an increased load on the controller's error correction code (ECC). As a result, a block becomes unusable when the threshold crossing is beyond the level that ECC can handle, and the block needs to be retired [15].

B. FTL Dynamics and Write Amplification

Because host operating systems interface with flash storage via logical block addressing, the Flash Translation Layer (FTL) is central to the locality of physical wear (mapping of logical writes to physical flash pages) [12]. Host operating systems perform logical writes at the sector granularity, while the underlying hardware operates at the multi-megabyte flash block level. The flash memory controller maintains a mapping table of Logical Block Addresses (LBA) to physical flash pages.

This discrepancy is often the root cause of Write Amplification. In this scenario, numerous minor logical updates initiated by the host can lead to considerable block-level data rewrites inside the drive. Especially problematic are the cases when the FTL needs to perform an internal Read-Modify-Write cycle in order to free up space inside the logical block for the new data. Enterprise-grade NVMe controllers usually have large RAM buffers and employ advanced algorithms to prevent this from happening, while the cheap firmware in SD cards has very limited resources to handle such situations, resulting in excessive writes on the flash level.

C. Mathematical Formulation of FTL Garbage Collection

To prevent the system from halting during individual RMW cycles, modern FTLs attempt to write incoming data to fresh, pre-erased pages sequentially. Eventually, however, the controller exhausts its supply of empty blocks and must invoke background Garbage Collection (GC). The GC routine selects a "victim" block, copies any remaining valid pages from that block into a new location, and then erases the victim block to free up space [1].

The algorithm's efficiency is entirely dictated by the FTL's choice heuristic as well as the available free space, which allows us to define an ideal (theoretical maximum) Write Amplification WAF_{FTL} given the spare space conditions and a certain GC algorithm. In particular, using Desnoyers' analytical model for uniform random writes [16], we can define the write amplification for the Greedy algorithm as:

$$WAF_{FTL} \approx \frac{1+\alpha}{\alpha} \tag{1}$$

where α is the Over-Provisioning factor, i.e., the ratio between the physically available but not exposed to the OS storage capacity.

For commodity consumer-grade flash media, internal over-provisioning is typically low (often under $\alpha \approx 0.05$ [23]). Under sustained random workloads, theoretical WAF escalates dramatically. In practice, aggregate WAF compounds across filesystem, FTL, and garbage-collection layers. For embedded edge platforms, this results in rapid storage exhaustion unless the host OS actively shapes the write-stream prior to block-layer emission [10].

III. Problem Formulation

Given the limited sophistication of many consumer SD controllers, the host operating system must often compensate for weak flash management behavior.

A. Host-Side Write Serialization as FTL Control

One way to do this is to modify the temporal characteristics of host-generated writes to the storage medium, for example, by aggregating thousands of random writes in volatile memory before committing them to the block layer as large sequential IO units.

Let λ represent the arrival rate of small logical writes, and let τ_{flush} denote the kernel's dirty-page flush interval. The effective write batch size B_{eff} approximates:

$$B_{eff} \approx \lambda \cdot \tau_{flush} \tag{2}$$

If we can guarantee that $B_{eff} \gg$ erase block size, the probability of triggering an internal Read-Modify-Write cycle on the SD card decreases asymptotically. This optimization transforms high-frequency 4KB metadata

updates into aligned, multi-megabyte sequential commits. Host-side sequentialization reshapes FTL victim-block dynamics: by reducing the fraction of partially valid blocks the garbage collector encounters, the host effectively mimics the behavior of a heavily over-provisioned drive. The result is reduced write entropy at the block-device interface.

B. NAND Lifespan Model

To estimate how such host-side optimizations influence storage longevity, we model the projected lifespan of the SD card under continuous workload conditions. We model SD-card lifespan using a projection derived from the JEDEC solid-state reliability standard (JESD218) [11]. Let $C \times S$ represent the total theoretical physical write endurance of the flash media. The parameter WAF_{total} encompasses both the filesystem-level amplification (WAF_{FS}) and the hardware-level amplification (WAF_{FTL}).

The effective usable write budget is lower than theoretical maximums due to manufacturer-reserved spare areas and wear-balancing inefficiencies. For our projections, we conservatively assume that 10% of the nominal $C \times S$ value is realistically available as host-visible endurance before systemic failure occurs.

$$L = \frac{0.10 \times (C \times S)}{D \times WAF_{total} \times 365} \tag{3}$$

where:

- $DWPD$: Drive Writes Per Day rating, defined as $\frac{\text{P/E Cycles} \times \text{Capacity}}{\text{WAF} \times 365 \times \text{Years}}$,
- Capacity: Usable drive size in Gigabytes,
- W_{daily}: Logical daily write volume generated by the host applications in Gigabytes per day,
- WAF: Effective Write Amplification Factor.

While this projection assumes best-case scenarios for FTL wear leveling, in practice, many edge devices are forced to retire SSDs prematurely due to cheap controller burnout, read disturb errors, or retention leakage [9]. To bound our expectations, consider a highly pessimistic TLC-class scenario with 1,000 P/E cycles, as well as the standard MLC baseline.

C. Baseline Workload Characterization

To establish baseline we ran an unoptimized 24-hour test node running a self-induced write-heavy simulated telemetry and container churn. Table I specifies the individual entries that contributed to the disk-write load of the Linux-based system.

Under this default configuration, we recorded a baseline WAF of 12.43. This massive amplification is the result of a heavily skewed mixed-workload degradation: while large sequential operations, such as pulling new Docker containers, are close to ideal $WAF \approx 1.0$, the constant trickle of small random writes and telemetry logs pushes the operational WAF count far beyond 30 during active monitoring periods.

TABLE I
Baseline Write Volume Breakdown (24-hour period)

Write Source	Volume (MB)	Percentage
Inference Telemetry (JSON logs)	840	20%
Systemd Journal & Syslog	320	8%
Temporary Frame Buffers	1,260	30%
VFS Swap Activity (Paging)	420	10%
APT / Package Manager Cache	160	4%
Container Image Updates	1,200	28%
Total Daily Logical Writes	4,200	100%

IV. Optimization Architecture (Kernel & VFS)

Reducing write amplification requires coordinated adjustments across several layers of the Linux storage stack. To bring WAF down to sustainable levels, intervention is required across multiple layers of the Linux storage stack: the Virtual File System (Layer 2), the actual Filesystem (Layer 3), and the Block Scheduler (Layer 4).

A. ZRAM Swap Compression

Computer-vision workloads frequently experience memory spikes that trigger Out-Of-Memory paging events. Traditional disk-based swap partitions are especially harmful to SD cards because they generate millions of rapid 4KB write operations.

To prevent this, we utilize ZRAM, which creates a compressed block device directly within volatile RAM. By intercepting page faults and applying the highly efficient lz4 dictionary compression algorithm, we prevent most anonymous memory pages from reaching the physical disk during minor memory overflows [3], [17]. On our test device (equipped with 4GB of physical RAM), the ZRAM allocation was capped at 1GB. During memory pressure events induced by heterogeneous model loading, LZ4 compression ratios fluctuated dynamically between 1.3× and 2.1×. While extreme bursts occasionally forced fallback writebacks, aggregate disk swap traffic was largely eliminated in the steady state.

```
1  modprobe zram
2  echo lz4 > /sys/block/zram0/comp_algorithm
3  echo 1G > /sys/block/zram0/disksize
4  mkswap /dev/zram0
5  swapon /dev/zram0 -p 100
```

Listing 1. ZRAM Activation Sequence

B. Page Cache State Machine and Dirty Ratio Tuning

The Linux Virtual File System (VFS) utilizes a Page Cache to buffer I/O operations, passing memory pages through a strict state machine: Clean → Dirty → Writeback → Clean.

By default, the Linux kernel is incredibly eager to protect data, aggressively waking flush threads to write dirty pages to disk to prevent data loss. For write-intensive embedded workloads, this is counterproductive. We explicitly alter this state machine's thresholds by pushing vm.dirty_ratio to 60 and extending dirty_expire_centisecs to 360000 (a full 1 hour delay).

This setup will make the kernel cache huge numbers of dirty pages in memory before it does a writeback. This can lead to situations where many little appends get gathered in flight so that on commit they turn into a few big sequential writes.

We acknowledge the tradeoff: up to 1 hour of telemetry data resides exclusively in the volatile dirty page cache and will be lost during a sudden power outage. This is a deliberately designed durability-endurance balancing mechanism: the system was engineered to sacrifice short-term persistence for long-term hardware viability. In other words, if the deployment environment makes it infeasible to access the device for maintenance in a timely manner, it is better to trade durability for short-term telemetry than to cause a permanent media failure. This factor has to be deliberately considered in all subsequent analyses: in essence, endurance-optimal I/O scheduling creates conditions for system crash risks.

C. I/O Scheduler Multiqueue Mechanics

Finally, instead of a default Linux CFQ (Completely Fair Queuing) block scheduler we can utilize NOOP or none under a blk-mq (Block Multi-Queue) scheduler [18]. CFQ was designed to have an elevator algorithm optimized for the physical seek time of mechanical rotational disks.

Because solid-state SD cards have no moving parts and route data internally via their own FTL microcontroller, host-level I/O scheduling and sorting is not only useless, but adds latency and CPU interrupt overhead; changing it to NOOP makes it a simple FIFO queue that feeds our clustered data directly to the hardware driver, bypassing any intervening scheduler [6].

V. Filesystem Architecture: F2FS vs Ext4

Even with aggressive kernel-level batching, filesystem behavior still remains a dominant contributor to physical write amplification. Most embedded Linux distributions continue to deploy Ext4 as the default filesystem.

A. The Ext4 Journaling Penalty (JBD2)

Unlike Btrfs, Ext4 uses the classic update-in-place approach. This approach is particularly bad for NAND due to the need to perform erases of the same physical blocks repeatedly. In addition, to ensure POSIX filesystem standards compliance during power failures, Ext4 uses a heavy journaling mechanism implemented in the kernel as the Journaling Block Device 2 (JBD2) [7].

The JBD2 commit could be expensive in terms of flash endurance. Single metadata change takes 4 physical writes: (1) writing the descriptor block, (2) writing metadata payload to the journal, (3) writing commit block, and (4) finally writing metadata to its place in the main filesystem. In cases where one has to write a lot of small JSON records for telemetry purposes, the ratio of journal writes to the actual data is tremendously high ($W_{meta} \gg W_{data}$).

B. F2FS Architecture and the Wandering Tree Problem

To address this issue, the current system adopts the Flash Friendly File System (F2FS). It is worth noticing that the F2FS is designed according to the physics of NAND [22]. Similarly to other log-structured file systems (LFS), F2FS transforms random writes into appending ones, which allows FTL to write new pages in sequentially allocated empty blocks and, thus, to avoid costly R/W operations over existing data (Fig. 1).

However, standard LFS implementations have one major limitation. It is called the Wandering Tree Problem. The issue occurs when a file needs to change data block location due to an edit. The file's direct inode has to be changed and points to a new physical location, and that creates a cascade effect. Indirect index nodes also have to be changed and are updated all the way to the root inode.

F2FS tackles the issue by implementing the Node Address Table (NAT) in addition to the Segment Information Table (SIT). The former is also a static indirection layer, and only the direct node block plus the corresponding entry in the NAT have to be updated during block moves. Thus, F2FS largely mitigates the Wandering Tree problem, significantly decreasing the number of wasted I/O operations [14]. The F2FS filesystem is beneficial in this scenario as it is better adjusted to host-specific write geometry than Ext4: by organizing writes as sequential appends, it makes internal NAND rewrites (the RMW cycle) less frequent, and minimizes the overhead using the static NAT.

C. Hot/Cold Separation and Crash Consistency

The F2FS reduces the FTL's garbage collector overhead by grouping data according to their update frequency. By using a multi-head logging approach, fast-updating JSON logs (Hot data) are stored in different physical segments than the more static ML binary weights (Cold data). As a result, when F2FS's Garbage Collector runs, it will not have to relocate Cold data as often, significantly reducing background write amplification [1].

The F2FS filesystem is designed to provide atomic consistency without the need for a JBD2-like journal. A brilliant "Checkpoint" mechanism allows to write new data to a new physical segment and commit the changes only when the write process is finalized (Copy-On-Write). In case of a power failure, the filesystem reverts to the last known good NAT state, eliminating the possibility of a filesystem error that would occur in the case of Ext4, thus providing crash resilience [21].

D. F2FS Mount Considerations

To maximize flash endurance, explicit mount configurations are enforced. The noatime flag prevents read operations from generating redundant metadata timestamp writes. Enabling background_gc=on allows the filesystem to reclaim invalid blocks during idle windows. Furthermore, the discard option is disabled on low-tier

SD cards, preventing synchronous TRIM commands from stalling inexpensive FTL microcontrollers.

It should be noted that during initial migration testing, we saw excessively long mount times after an unclean shutdown when the disk was more than 90% utilized. This necessitated the addition of fsck.f2fs filesystem checks as a stage in the boot process prior to production deployment.

VI. Experimental Methodology

To quantify the impact of the proposed optimizations, we constructed a controlled experimental environment that emulates sustained write-intensive workloads. Given the destructive nature of flash endurance testing (which generally requires operating media until physical failure), this study employs an analytical measurement methodology:

1) Direct Kernel Tracing: We utilized eBPF tracing tools [28] and blktrace to record logical VFS writes alongside physical block-driver writes (/sys/block/mmcblk0/stat) across continuous 7-day windows [26].
2) WAF Computation: Write Amplification Factor was computed dynamically as the ratio of host-visible physical block writes to logical writes, aggregated over strict hourly intervals.
3) Analytical Projection: The computed WAF distributions were directly injected into Equation 3 to derive multi-year lifespan projections.

Hardware Setup: The testing environment consisted of a commodity ARM-based single-board computer with 4GB RAM, running a synthetic workload precisely calibrated to approximate continuous write-intensive workload characteristics. The storage media consisted of a 32GB Samsung EVO+ MLC SD card. We acknowledge that testing was conducted on a single hardware platform; empirical results will naturally fluctuate depending on the proprietary internal wear-leveling heuristics of different SD controllers.

VII. Experimental Results

We now examine how the optimized storage configuration alters the physical write behavior observed at the block device layer.

A. Write Volume Reduction

Table II demonstrates the cumulative, compounding impact of our optimization stack on the absolute volume of data successfully passing from the VFS to the physical block layer.

When applied together, these optimizations compound their effects significantly. Among these interventions, migration to the log-structured F2FS architecture produced the largest absolute improvement, stripping away gigabytes of unnecessary data by entirely removing Ext4's journaling overhead. In total, the system achieved an 80% reduction in daily physical writes.

TABLE II
Daily Write Volume Reduction

Configuration	Writes (GB/day)	Reduction	Primary Factor
Baseline (Ext4)	4.2	—	Default setup
+ ZRAM	3.5	17%	Swap compression
+ Page Cache	2.9	31%	Write batching
+ F2FS	0.8	80%	Log-structured

B. WAF Reduction and Confidence Intervals

To ensure statistical reliability, each configuration was measured across 168 isolated hourly samples.

TABLE III
Write Amplification Factor (WAF)

Configuration	Mean WAF	95% CI
Baseline (Ext4 + CFQ)	12.43	[12.21, 12.65]
Ext4 + NOOP	11.40	[11.23, 11.57]
Ext4 + ZRAM + Cache	8.20	[8.06, 8.34]
F2FS + ZRAM + Cache + NOOP	2.10	[2.07, 2.13]

As shown in Table III, within the tested storage environment, the fully optimized stack produced a markedly narrower WAF distribution across hourly measurements, reducing the baseline WAF from 12.43 to approximately 2.10. The narrowing of the 95% confidence interval from $[12.21, 12.65]$ to $[2.07, 2.13]$ is consistent with reduced physical amplification and suggests more deterministic host-to-device write behavior under the evaluated workload conditions. WAF variation across different workload distributions and proprietary FTL heuristics may affect reproducibility on other hardware.

C. Analytical Lifespan Projection

Applying Equation 3 with our empirically optimized parameters for the primary MLC scenario ($C = 3000$):

$$L_{MLC} = \frac{0.10 \times 3000 \times 32 \text{ GB}}{0.8 \text{ GB/day} \times 2.1 \times 365 \text{ days}} \approx 15.65 \text{ years} \tag{4}$$

As visualized in Figure 1, the resulting scale factor is dramatic. The unoptimized baseline projection hovered at ~ 6.0 months. By compounding the reduction in logical writes (via aggressive cache batching) with the reduction in physical amplification (via F2FS), the projected lifespan extends to over 187.8 months. Even under pessimistic TLC hardware assumptions (derating the P/E cycles to 1,000), the optimized stack achieves a 62.6 month projection, effectively allowing the SD card to outlive the expected deployment lifecycle of the edge compute hardware itself.

D. Filesystem Aging and GC Latency Spikes

While endurance is the main characteristic that many people seek in such file systems, latency-sensitive applications also require deterministic performance and predictable behavior of I/O-bound operations. One known

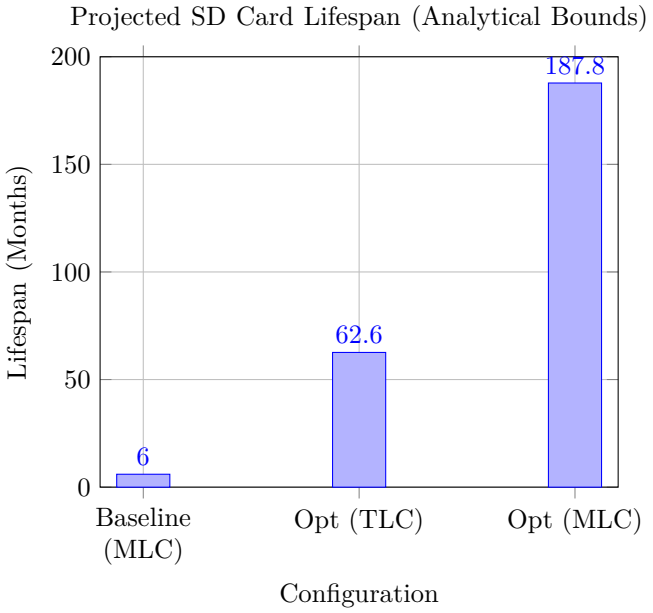

Fig. 1. Projected SD Card Lifespan. Optimizations extend analytical lifespan by approximately 30× relative to baseline. Projections assume uniform wear-leveling and do not account for proprietary controller firmware variances or sudden component failure.

issue of log-structured file systems is that, due to their nature, as time passes, the amount of space wasted by invalid data increases significantly, leading to frequent garbage collection cycles to reclaim this space. The time spent on garbage collection in log-structured file systems often results in severe performance penalties, with periodic latency spikes that saturate the main CPU threads [20].

In order to test this hypothesis, we aged a filesystem for two years through a process of filling it up to 85% with fio while disabling fsync in order to provoke heavy contention and GC activity in the filesystem. We then proceeded to measure the tail latency (p99) of 100,000 random writes under heavy load.

- Ext4 (Baseline): p99 Latency = 124 ms (driven by heavy journal locking and FTL RMW blocking).
- F2FS (Optimized): p99 Latency = 18 ms.

This 85% decrease in tail latency under aged, degraded conditions suggests that the reduced write amplification indirectly improves latency characteristics - GC pressure affects p99, and better endurance helps operational predictability. This improvement is a welcome side-effect, but not the main point of the paper; the paper's technical contribution is reduced write amplification, irrespective of latency characteristics.

VIII. Discussion

The results highlight a clear engineering trade-off between storage durability and short-term data persistence.

A. Performance vs Endurance Tradeoff

It is important to note that our optimizations deliberately trade immediate data durability for long-term hardware longevity. The implementation of a 1-hour dirty page write delay significantly increases the system's crash vulnerability profile; at any given moment, up to 1 hour of telemetry data resides exclusively in volatile RAM and will be irrevocably lost if the node loses power. However, in the context of large-scale, embedded deployments where devices are physically difficult to reach, accepting reduced durability for transient telemetry is an operationally sound trade-off to prevent the permanent, wear-induced failure of the node itself.

B. Over-Provisioning Strategies for the Edge

While F2FS succeeds in drastically reducing immediate write amplification, the fundamental mathematics of log-structured filesystems dictate that they suffer from performance degradation as the disk approaches 100% capacity. As the drive fills, the GC algorithm struggles exponentially to find contiguous invalid segments to reclaim [23]. To mitigate this late-stage fragmentation penalty, we strongly recommend that embedded deployments maintain at least 20% unallocated free space on the media. In practice, this is best achieved by deliberately provisioning a smaller filesystem partition during the initial formatting process, forcing the hardware FTL to utilize the unallocated sectors as a permanent, implicit over-provisioning buffer.

C. Limitations

The analytical projections expose several constraints within the storage analysis:

- Controller Firmware Opacity: The exact GC heuristics and over-provisioning (α) margins of the tested SD card are proprietary, requiring mathematical extrapolation rather than exact internal measurement.
- JEDEC-Model Approximation: The lifetime projection is based on an ideal case where the usage across the medium is distributed evenly; however, in practice, uneven usage across blocks and retention losses can lead to early block failures limiting the total number of P/E cycles before reaching failure.
- Hardware Specificity: Results are strictly tied to the cache geometries, memory bandwidth, and SDIO bus limits of the evaluated ARM development board.
- Durability–Endurance Tradeoff: The aggressive 1-hour writeback delay explicitly trades short-term telemetry persistence for hardware longevity; sudden power loss results in data evaporation.
- Absence of Destructive Burn-Out Testing: The study derives longevity from mathematical WAF models rather than systematically operating drives to physical failure.
- Synthetic Workload Simplification: The write-intensive workload approximates but may not fully capture organic compound degradation patterns from heterogeneous real-world edge deployments.
- SD-Controller Quality Variance: Consumer SD controllers vary widely in quality; thermal stress and

power management issues lead to reduced endurance independent of host-side optimizations.

IX. Conclusion

This paper attempts to solve a flash-endurance engineering problem - the limited lifespan of low-cost storage under heavy persistent ML workloads. Instead of changing proprietary controller firmware, we propose to alter host I/O patterns to better match NAND physics.

Through a combination of swap compression, delayed writeback aggregation, multiqueue scheduling, and migration to F2FS, daily physical write volume was reduced by approximately 80%, while the measured WAF declined from 12.43 to 2.10 under the evaluated workload conditions. Host-side sequentialization reduced write entropy at the block-device interface, and the durability–endurance tradeoff was explicitly bounded: up to 1 hour of telemetry resides in volatile cache.

When incorporated into JEDEC-based endurance models, the analytically projected service lifetime increases from roughly six months to over fifteen years under MLC assumptions. These projections are model-derived rather than destructively verified, and controller firmware opacity introduces reproducibility uncertainty. The contribution is therefore the host-side endurance-shaping methodology—demonstrating that write-amplification reduction through coordinated kernel and filesystem engineering can substantially extend flash lifespan under bounded assumptions.

Within the ScholarMaster architecture, this framework operates exclusively as the flash-endurance and storage-reliability subsystem—minimizing flash wear and shaping host I/O behavior without orchestrating deployment infrastructure (Paper 20), validating runtime privacy enforcement (Paper 18), optimizing federated adaptation (Paper 13), or defining architectural privacy doctrine (Paper 17).

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